

Static copy of an interactive page. The sliders, buttons and animations work only in the live version: use the link in the Canvas announcement. Numbers shown here are for $T = 2, J = +1, h = 0$.

DSAA 4011 • LABORATORY 3 • INTERACTIVE EDITION

Sixteen spins, 65,536 configurations, every number exact

- **Model:** $p_T(s) = e^{-E(s)/T} / Z_T$ on a 4×4 grid of spins
- **Method:** your browser enumerates all 2^{16} configurations
- **Every number is exact** until Station 8
- Stations 1–8: **explore**
- Stations 9–10: **build your own**
- Station 11: **research questions**
- Move with the arrow keys or the buttons below

Touch the model

- $E(s) = -J \sum s_i s_j - h \sum s_i$ over 24 neighbouring pairs
- **Aligned pair:** energy $-J$
- **Broken pair:** energy $+J$

All sixteen spins are +1 ($J = 1$, $h = 0$). You flip one *corner* spin. By how much does the energy rise?

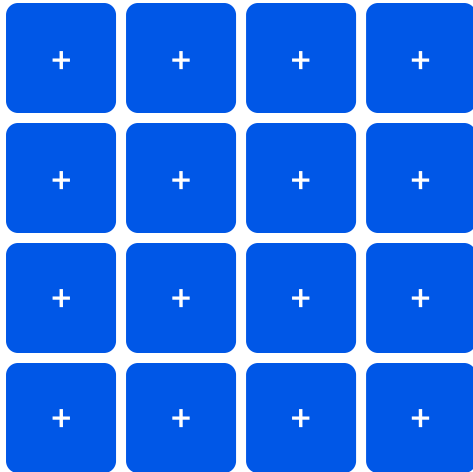
2

4

8

16

Click a spin to flip it



■ +1 ■ -1

All +1

Stripes

Checkerboard

Random

This configuration

Broken pairs	0 of 24
Energy $E(s)$	-24.00
Magnetisation M	16
Ratio to the most probable state	1.000e+0
Exact probability $p_T(s)$	8.227e-2

- **Ratio:** two energies are enough
- **Probability:** needs Z_T , a sum over 65,536 configurations
- Slide T and watch both

In today's ML

- A language model's **logit is a negative energy**
- **Nobel Prize in Physics 2024:** Hopfield and Hinton
- Hopfield network, Boltzmann machine = Ising models with **learned couplings**

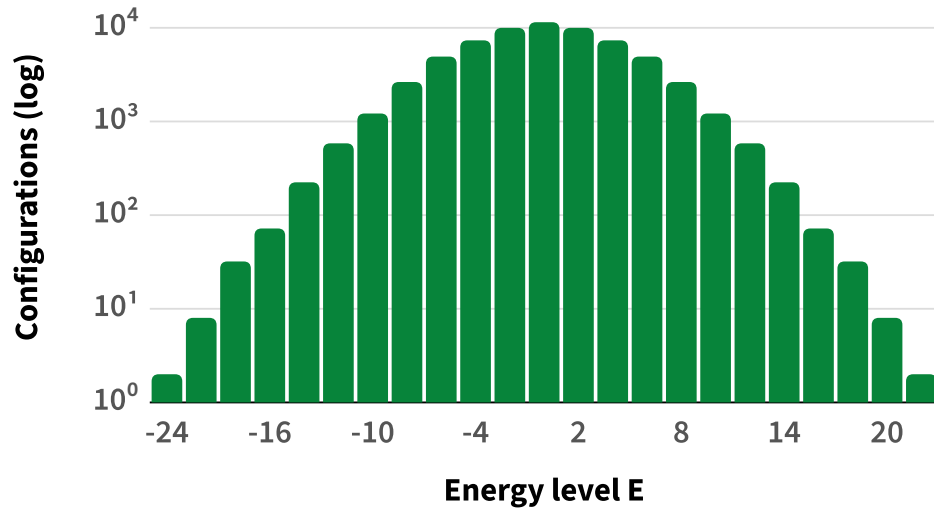
Count the configurations at each energy

- Level probability = **count** × **weight**
 - **Count**: the entropy side
 - **Weight**: the energy side
-

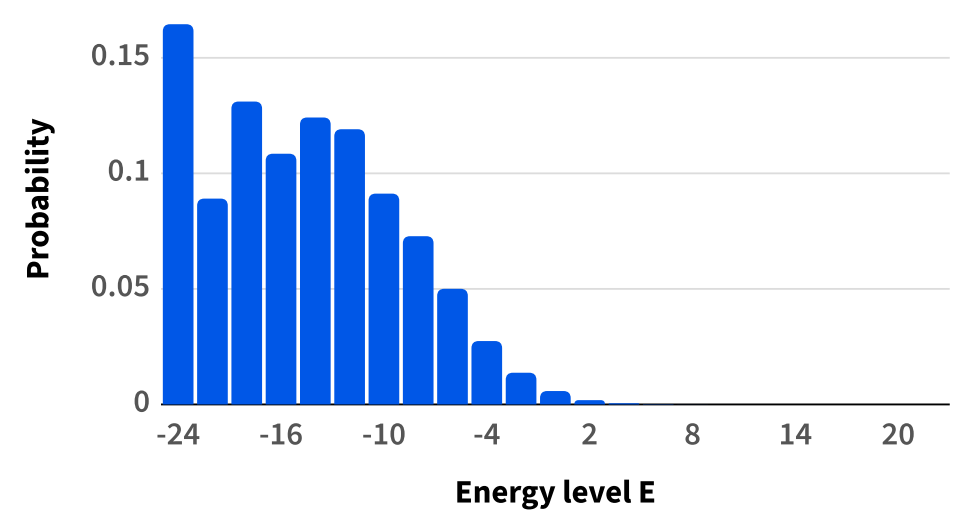
How many of the 65,536 configurations reach the lowest energy, -24?

☐ 1☐ 2☐ 16☐ 32

Number of configurations per energy level (log scale, $J = 1$, $h = 0$)



Probability of each level at the current T



Most probable level: -24, probability 0.1645. Above $T \approx 2.1$ the ground level stops being the most probable level.

In today's ML

- At $T = 3$: one ground state 0.0110, the most probable single state
- Level -8: 2,638 states, total 0.1401; ground level: only 0.0220
- **The most probable state is not where the probability is**
- Large language model (LLM): max-probability text is short, generic, repetitive; **sampling gives typical, human-like text** (Holtzman et al., ICLR 2020)

A sum that overflows

- At $T = 0.01$ the largest term is e^{2400}
- Double precision stops near e^{709}**

You compute Z by adding up $e^{-E/T}$ directly at $T = 0.01$. What does the program return?

☐ A very large number

☐ inf

☐ 0

☐ An exception

Two ways to compute $\ln Z$, at a temperature you choose

Station temperature  0.010

Largest exponent $a_{\max} = 24/T$ 2400.0

Direct sum $\sum e^a$ inf

ln of the direct sum inf

Log-sum-exp 2400.6931

The three lines that fix it

```
a = -E / T
m = a.max()
lnZ = m + log(sum(exp(a - m)))
```

- Subtract the maximum: largest term becomes $e^0 = 1$
- Nothing overflows**
- Add m back: exact

In today's ML

- Every softmax and cross-entropy** in PyTorch and JAX uses this trick
- LLM training loss = log-sum-exp over about 100,000 tokens

Sweep the temperature

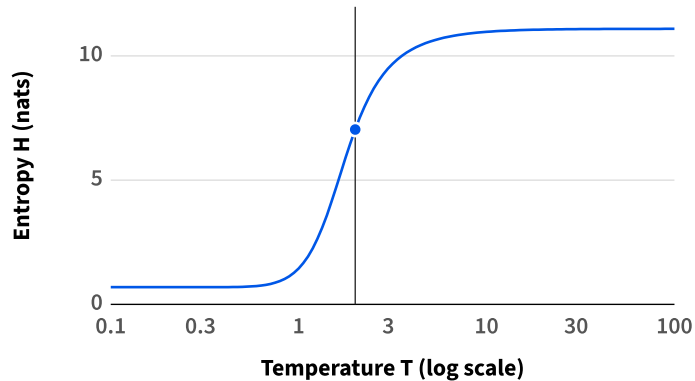
- Entropy H and probability of the ground states
- Heat capacity: $C = dU/dT$, how fast the mean energy $U = E[E]$ rises with T
- Identity: $C = \text{Var}(E)/T^2$, so C is large where the energy **fluctuates most**
- Per spin: $C/16$, comparable across lattice sizes. **All exact**

As $T \rightarrow 0$ ($J = 1, h = 0$), the entropy of the grid tends to...

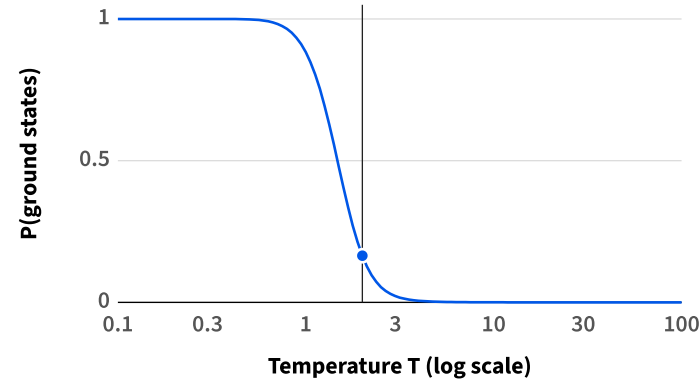
0

 $\ln 2 \approx 0.69$ $16 \ln 2 \approx 11.09$

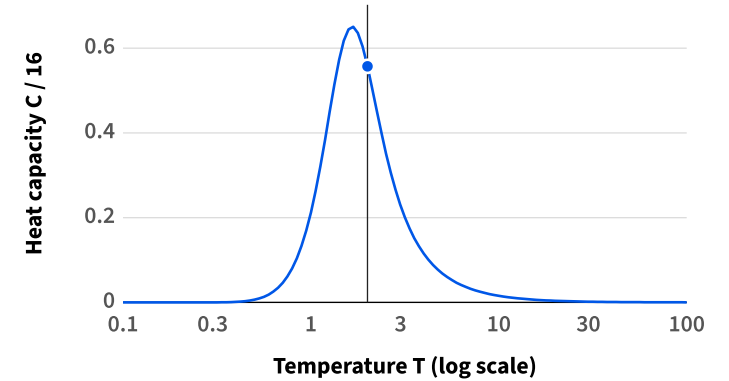
Entropy H (nats)



Probability of the ground states



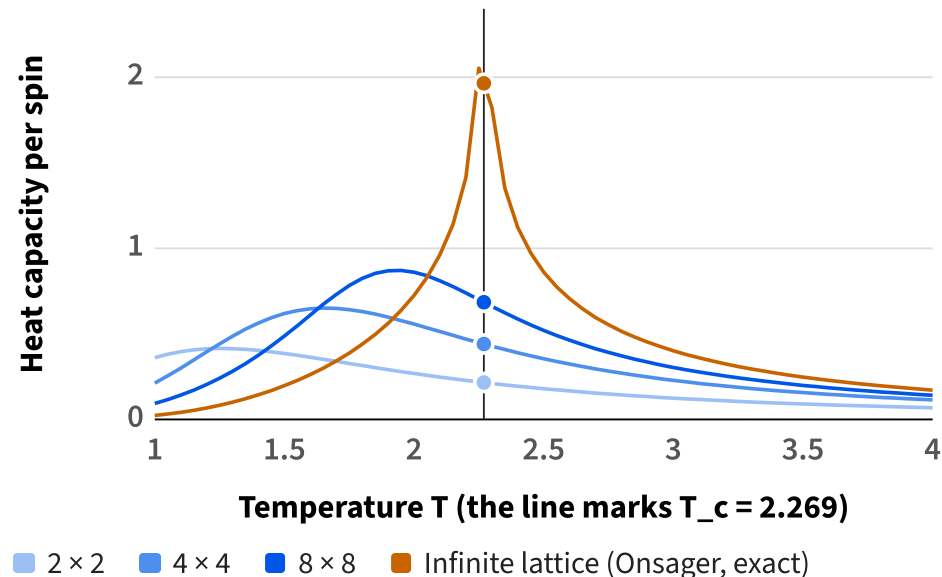
Heat capacity per spin



$\ln Z$	14.4977	H	7.0382
$P(\text{ground})$	0.1645	$C/16$	0.5573

- Peak = **where the distribution changes fastest**, $T \approx 1.66$ on this 4×4 grid

Towards the phase transition: heat capacity per spin as the lattice grows



What the figure shows

- Peak position: $1.26 \rightarrow 1.66 \rightarrow 1.93 \rightarrow T_c = 2/\ln(1 + \sqrt{2}) = 2.269$
- Peak height: $0.41 \rightarrow 0.65 \rightarrow 0.87 \rightarrow \text{infinite}$ (a logarithmic divergence)
- **Phase transition:** below T_c the infinite lattice stays magnetised; above it, it does not
- A finite grid has only a smooth peak: **no true transition**
- Reference: L. Onsager, Physical Review 65, 117–149 (1944); T_c first located by Kramers and Wannier, Physical Review 60, 252–262 (1941)

In today's ML

- **LLM temperature** is this T : logits $\div T$ before the softmax
- **Overconfident classifier**: it says “90% sure” but is right only 70% of the time
- **Temperature scaling**: divide all logits by one number $T > 1$, chosen on held-out data to minimise the log loss
- The ranking, and so the accuracy, is unchanged; **the stated confidence drops to match reality** (Guo et al., ICML 2017)

The mean hides two modes

- **Magnetisation:** $M = \sum s_i$, from -16 to +16

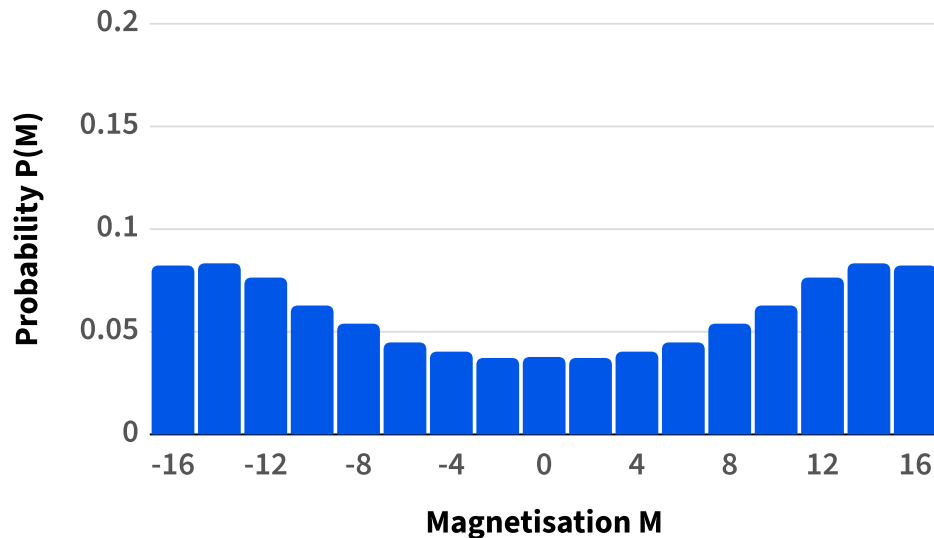
At $T = 1.5$ with $h = 0$, almost half the probability sits on the two fully aligned states. What is the mean of a single spin, $E[s_0]$?

About 0.83

About 0.48

Exactly 0

Distribution of the magnetisation M



Summaries

Mean spin $E[M]/16$	0.0000
Mean alignment $E[M]/16$	0.6205
$P(M = 16)$	0.1645
$P(M = 0)$	0.0377

- **Mean spin:** average of $M/16$, keeps the sign; +1 and -1 modes cancel
- **Mean alignment:** average of $|M|/16$, ignores the sign; 1 = all spins agree, near 0 = half up, half down
- Move the field h to 0.10
- **One mode wins:** the mean spin jumps
- The coupling did not change

In today's ML

- Predicting the **mean of two modes** gives something that never occurs
- So generators **sample** instead of averaging
- **One summary statistic cannot certify a generative model**

Correlation on a grid is not correlation on a chain

- **Chain:** correlation at distance d is $\tanh(J/T)^d$
 - **One path;** every link weakens it
-

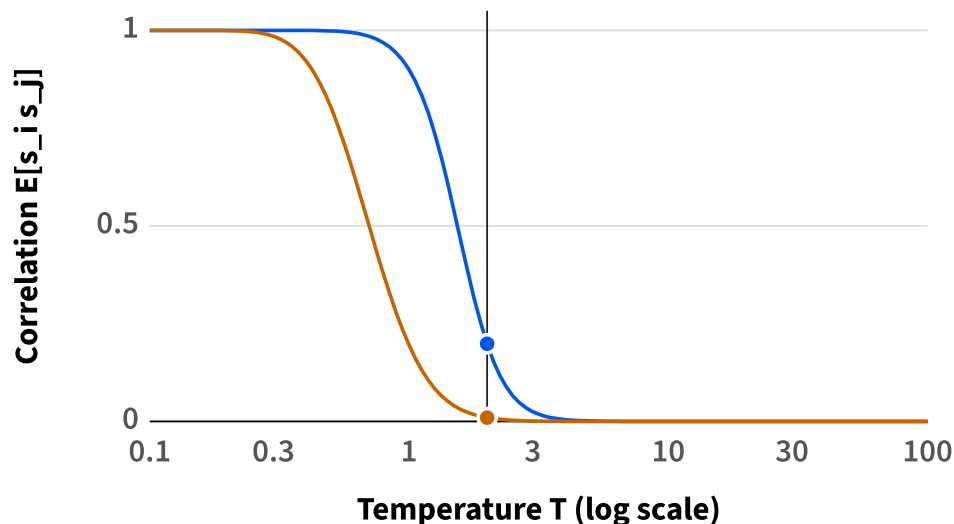
Opposite corners of the grid are 6 edges apart. At $T = 2$ the chain formula gives 0.0097. The exact grid correlation is...

Smaller than 0.0097

About the same

About 20 times larger

Correlation against temperature



Pick two spins

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Distance d

6

$E[s_i s_j]$ on the grid

0.1985

Chain formula

0.0097

Mutual information (nats)

0.0198

- Try $J = -1$: neighbours disagree
- Corners stay positive: **six sign flips cancel**

In today's ML

- Many paths carry more than one
- Attention**: every token reaches every other in one step
- Attention update = retrieval step of a **modern Hopfield network** (Ramsauer et al., ICLR 2021)

The same temperature inside a language model

- Teaching example: “Please close the ____”
 - **Logits are negative energies**
 - $q_T(y) = e^{z_y/T} / \sum e^{z/T}$
-

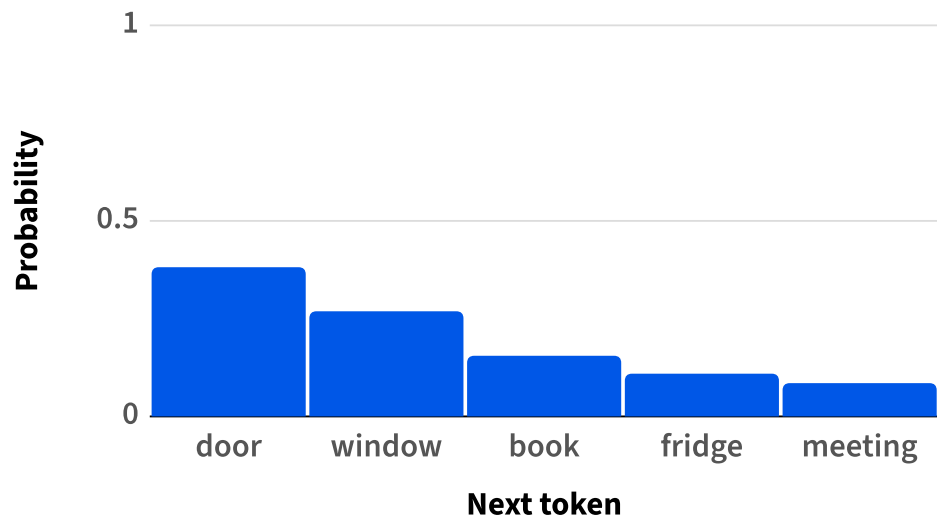
You raise the temperature from 1 to 5. The probability of the top token “door”...

Rises

Falls towards 1/5

Does not change

Next-token probabilities at the current T



Entropy (nats)

1.4616

Perplexity $\exp H$

4.31

Sample 40 continuations

Sample

- **Low T :** every draw is “door” (greedy)
- **High T :** five tokens near 20% each, perplexity 5

In today's ML

- **Alignment is a Boltzmann distribution:** $\pi(y) \propto \pi_{\text{ref}}(y) e^{r(y)/\beta}$
- Kullback–Leibler (KL) penalty strength β = temperature
- Basis of **Direct Preference Optimization (DPO)**, which tunes a model on human preference pairs without a separate reward model (Rafailov et al., NeurIPS 2023)

Sampling: build typical configurations one spin at a time

- **Problem:** on a big grid the exact sum over all configurations is impossible
 - **Idea:** replace the sum by an average. Exact: $E[g] = \sum_s p(s) g(s)$, 2^n terms, each needing Z
 - Sampling: draw $s_1, \dots, s_N$ with the right frequencies; then $E[g] \approx (1/N) \sum_t g(s_t)$. **No Z , no 2^n terms**
 - **One update:** pick a spin at random and look only at its neighbours
 - Local field $f = J \times (\text{sum of neighbours}) + h$; set the spin to +1 with probability $p = 1/(1 + e^{-2f/T})$
 - Example at $T = 2$: four neighbours +1 give $f = 4$, $p = 0.982$; two against two give $f = 0$, $p = 0.5$
 - **One sweep** = one update per spin on average. The estimate is the average over all sweeps so far
-

Does this update rule ever need the normaliser Z ?

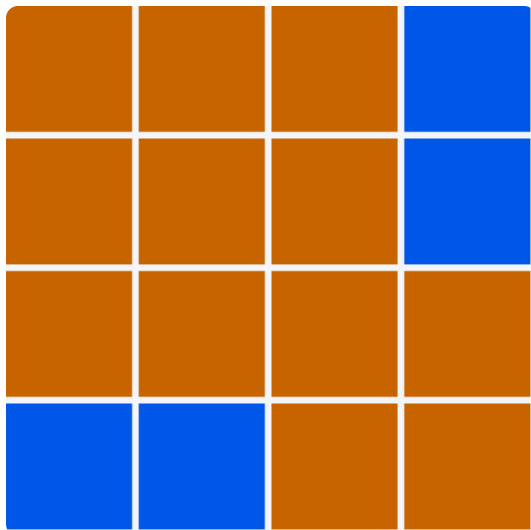
Yes, once per update

Yes, once at the start

No, never

The sampler (this rule is called Gibbs sampling)

Press “One update” to see a single step explained.



One update

Run

Reset

4 × 4

32 × 32

The average replaces the sum: check it against the exact value

Sweeps	0
Estimate of $E M /n$	–
Exact (4×4 only)	0.6205
Difference	–
Estimate of $E[s_5 s_6]$	–
Exact (4×4 only)	0.6776

- Here $g(s) = |M(s)|/n$: each sweep contributes one value, and the estimate is their mean
- Press Run on 4×4 : the difference shrinks, roughly like $1/\sqrt{N}$
- **This is how the impossible sum is solved:** not computed, but estimated to any accuracy you are willing to wait for
- Switch to 32×32 : 2^{1024} configurations
- **No exact value exists on any computer**
- **The sampler does not care**
- Lower T below 2: domains form

In today's ML

- Sampling without Z , by local changes only
- Same idea: **Markov chain Monte Carlo (MCMC), Langevin dynamics, diffusion models**
- Week 4: when does it converge?

Store one picture, damage it, watch it repair

- **So far:** the couplings were given. **Now the reverse:** choose couplings so that the network settles into a picture you pick
 - **Picture:** one configuration x of the 16 spins. Here it is the letter T
 - **Field** on spin i : $f_i = \sum_j W_{ij} s_j$, the weighted vote of the other 15 spins
 - **Stable:** every spin has the sign of its field, so no spin wants to flip
 - **To store a picture** = to choose W so that the picture is stable
 - **To repair** = flip a spin that disagrees with its field, and repeat until none does
-

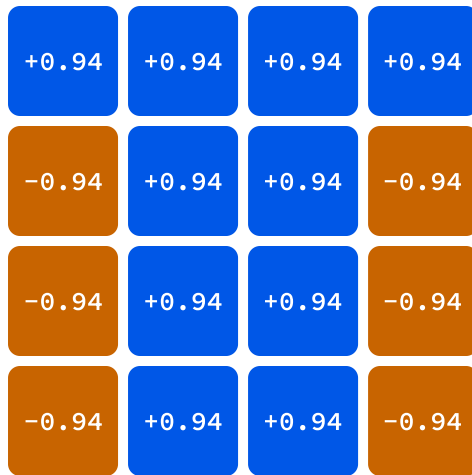
The T is stored. You flip one of its pixels. On its next update, that pixel...

Stays wrong

Flips back

Drags a neighbour with it

The network. Each cell shows the field on that spin



■ spin +1 ■ spin -1 ■ disagrees with its field

Repair one spin

Repair all

Damage 3 pixels

Back to the stored T

What is happening

Pixels that differ from the stored T

0 of 16

Spins that disagree with their field

0

Energy $E(s) = -\frac{1}{2} \sum W_{ij} s_i s_j$

-7.50

Click a pixel to damage the picture.

- **Couplings used here:** $W_{ij} = x_i x_j / 16$, from the T itself. Two pixels of the T that are equal: $+1/16$. Two that differ: $-1/16$. Station 10 derives this
- **Where ± 0.94 comes from:** each of the other 15 spins votes $W_{ij} x_j = x_i / 16$. Fifteen equal votes: $15 \times 1/16 = 15/16 = 0.9375$, shown as 0.94, with the sign of x_i
- **With k wrong pixels:** a wrong neighbour votes the other way, so a correct spin has field $(15 - 2k)/16$. Three wrong: $9/16 = 0.56$ on a correct spin, $11/16 = 0.69$ on a wrong one
- Stored T: every field agrees with its spin. **Energy -7.50, the bottom of the valley**
- Damage a few pixels, then repair one spin at a time: **each repair lowers the energy**
- Damage 9 or more: it repairs to the **negative picture**, which is stable too (the mirror image of Station 5)

In today's ML

- This is a **Hopfield network** (Hopfield, PNAS 1982): a Boltzmann machine whose valleys you design
- **Content-addressable memory**: you retrieve a whole item from a damaged part of it, not from an address
- The same retrieval idea, with a softmax in place of the sign, is transformer attention (Ramsauer et al., ICLR 2021)

Hebb's rule: how the couplings are chosen

- **Wanted:** for every spin, the field $f_i = \sum_j W_{ij}x_j$ has the sign of x_i . Then the picture is stable
- **Try** $W_{ij} = x_i x_j / n$. One vote is $W_{ij}x_j = x_i x_j x_j / n = x_i / n$, because $x_j x_j = 1$. **All 15 votes say x_i :** $f_i = 15/16 \cdot x_i$
- **Compute it:** multiply the two pixel values and divide by 16. T of Station 9: row 1 pixels are +1 and +1, so $W = (+1)(+1)/16 = \mathbf{+1/16}$. Row 1 column 1 is +1 and row 2 column 1 is -1, so $W = (+1)(-1)/16 = \mathbf{-1/16}$
- **Several pictures:** compute this for each picture and add. $W_{ij} = (1/n) \sum_{\mu} x_i^{\mu} x_j^{\mu}$. This is **Hebb's rule**
- **The price:** a picture now also receives the votes meant for the others. With too many pictures these extra votes win, and the pictures stop being stable

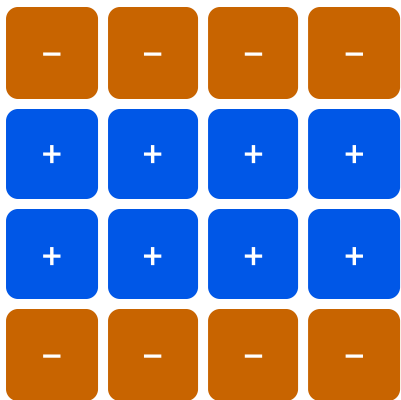
Two pixels are equal in picture 1 and different in picture 2. Hebb's rule gives their coupling as...

+1/8

0

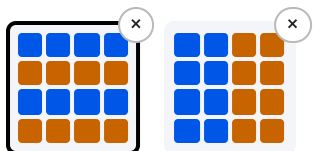
-1/8

Draw a picture, then store it



Store Random picture Draw the T Forget all

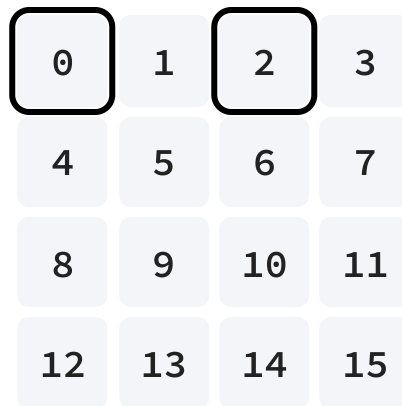
Stored patterns (2 of 10)



memory 1 memory 2

- Two example memories are stored to start: horizontal and vertical stripes
- Click x to forget a memory

Inspect one coupling: pick two pixels

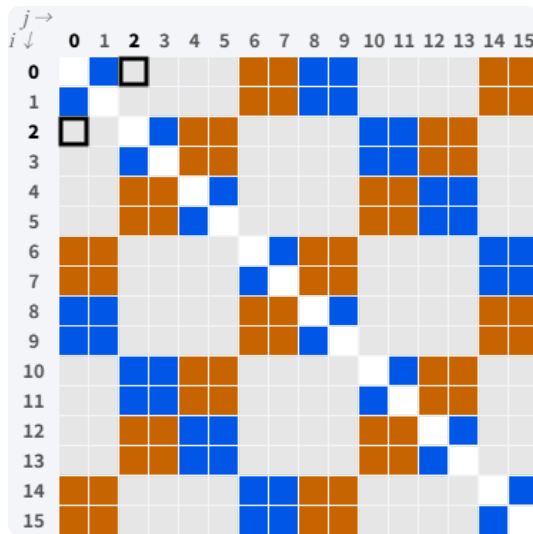


Memory **pixel 0** **pixel 2** **product**

picture 1	+1	+1	+1
picture 2	+1	-1	-1

$$W_{0,2} = (\text{sum of products})/16 = 0/16 = \mathbf{0.0000}$$

The coupling table: one square for every pair of pixels



- **How to read it:** pixels are numbered 0 to 15, row by row, as in the picker. The square in row i , column j is the coupling W_{ij} between pixel i and pixel j
- **Click any square** to load that pair into the inspector and see how its value was computed
- 16 pixels make $16 \times 15 / 2 = 120$ **different pairs**. Each pair appears twice, as (i, j) and (j, i) , so the table is symmetric. The white diagonal: a pixel has no coupling with itself
- ■ positive: want to agree ■ negative: want to differ ■ zero: no preference

What to try

- Two example pictures are stored: horizontal and vertical stripes. Pixels 0 and 2: equal in one, different in the other, coupling 0
- Forget all and store one picture: **every coupling is $\pm 1/16$**
- Store more pictures: couplings where the pictures disagree **fade to grey**. That fading is the pictures competing for the same couplings

In today's ML

- **Hebb's idea (1949)**, often summarised as “cells that fire together wire together”: a learning rule with no gradient and no error signal
- Each coupling is a **pair statistic of the data**, exactly the average $E[s_i s_j]$ that a Boltzmann machine is fitted to
- Gradient training of a Boltzmann machine starts from this rule and subtracts the model's own pair average

Three research questions you can start this week

- Tools: energy, temperature, entropy, KL divergence, sampler
- **Each first experiment fits in one afternoon**

Does entropy reveal when a language model is guessing?

- **Question:** invented answers disagree across samples. Does entropy flag them?
- **You have:** Station 7, sampling at T and entropy
- **First experiment:** ask one question 10 times at $T = 1$; group by meaning; compare entropy for right and wrong answers

Read: Farquhar, Kossen, Kuhn and Gal, “Detecting hallucinations in large language models using semantic entropy,” Nature 630, 2024.

Alignment as a Boltzmann distribution

- **Question:** $\pi \propto \pi_{\text{ref}} e^{r/\beta}$. How do reward and diversity trade off in β ?
- **You have:** the five-token model, entropy, KL divergence
- **First experiment:** choose rewards; for β from 0.1 to 10 compute reward, entropy, KL; plot reward against KL

Read: Rafailov et al., “Direct preference optimization: your language model is secretly a reward model,” NeurIPS 2023.

Generating data from an energy you cannot normalise

- **Question:** can sampling without Z generate images or molecules?
- **You have:** the ratio trick and a sampler with exact checks
- **First experiment:** sweeps needed to reach exact $E[M]$ within 0.01, against T ; explain the slow region with Station 5

Read: Song and Ermon, “Generative modeling by estimating gradients of the data distribution,” NeurIPS 2019. Ho, Jain and Abbeel, “Denoising diffusion probabilistic models,” NeurIPS 2020.

From question to project

- Write the question, the smallest experiment, and what would change your mind
- Bring it to office hours
- **A careful negative result on a small model is a real contribution**

Evidence card

- **Claim:** one thing you now believe
- **Test:** the station and number that support it
- **Limit:** what the test does not show

Claim

Example: a zero mean does not imply weak structure.

Test

Which station, which settings, which number?

Limit

What would this test not show?

No predictions committed yet.

Download the card

Clear my predictions

Every number is computed in your browser by enumeration and matches the course program `labs/week03/ising.py` to four decimals. Station 7 uses constructed logits.

References. C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, “On calibration of modern neural networks,” ICML 2017. J. J. Hopfield, “Neural networks and physical systems with emergent collective computational abilities,” PNAS 79, 2554–2558, 1982. D. O. Hebb, The Organization of Behavior, Wiley, 1949. H. Ramsauer et al., “Hopfield networks is all you need,” ICLR 2021. R. Rafailov et al., “Direct preference optimization,” NeurIPS 2023. L. Onsager, “Crystal statistics I. A two-dimensional model with an order-disorder transition,” Physical Review 65, 117–149, 1944. H. A. Kramers and G. H. Wannier, “Statistics of the two-dimensional ferromagnet,” Physical Review 60, 252–262, 1941. A. Holtzman, J. Buys, L. Du, M. Forbes and Y. Choi, “The curious case of neural text degeneration,” ICLR 2020. S. Farquhar, J. Kossen, L. Kuhn and Y. Gal, “Detecting hallucinations in large language models using semantic entropy,” Nature 630, 2024. Y. Song and S. Ermon, “Generative modeling by estimating gradients of the data distribution,” NeurIPS 2019. J. Ho, A. Jain and P. Abbeel, “Denoising diffusion probabilistic models,” NeurIPS 2020.